project_rcode_final

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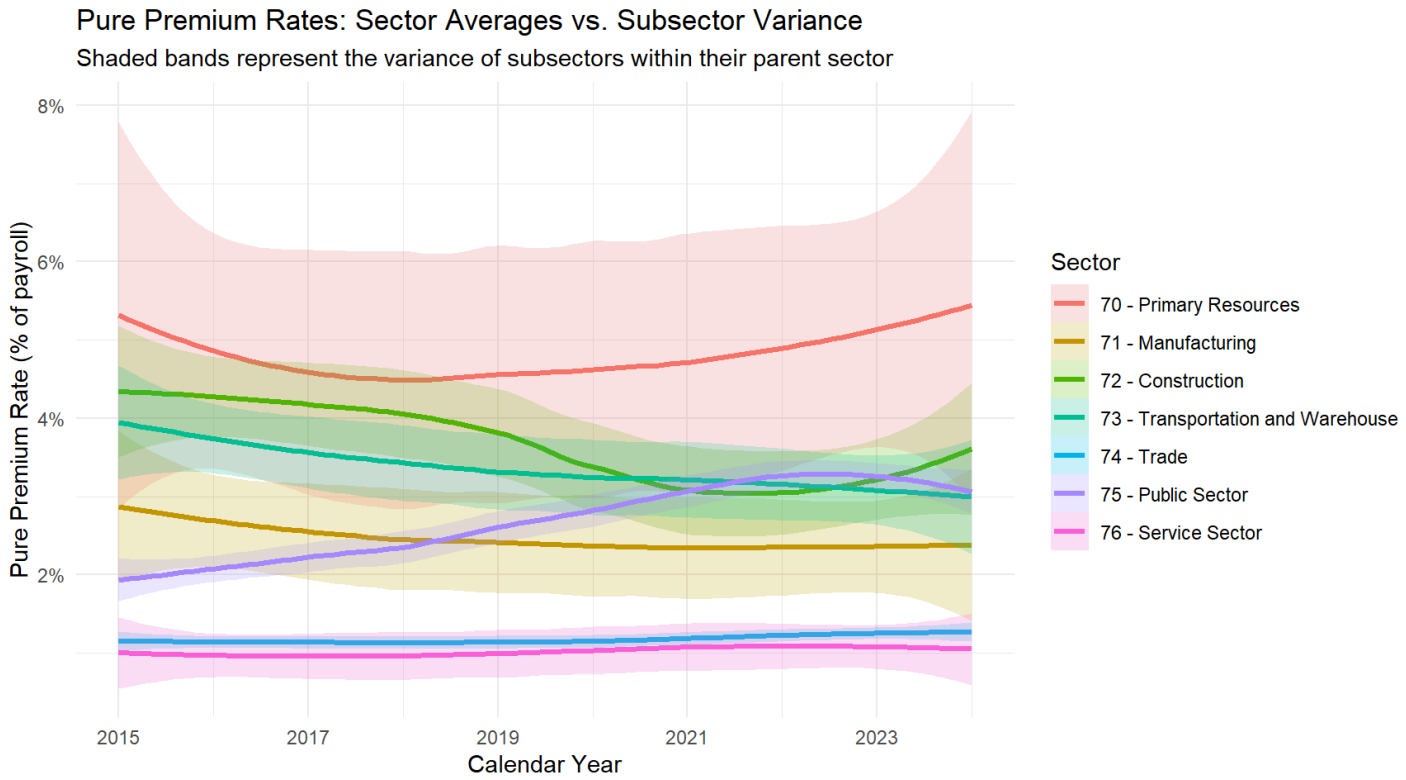
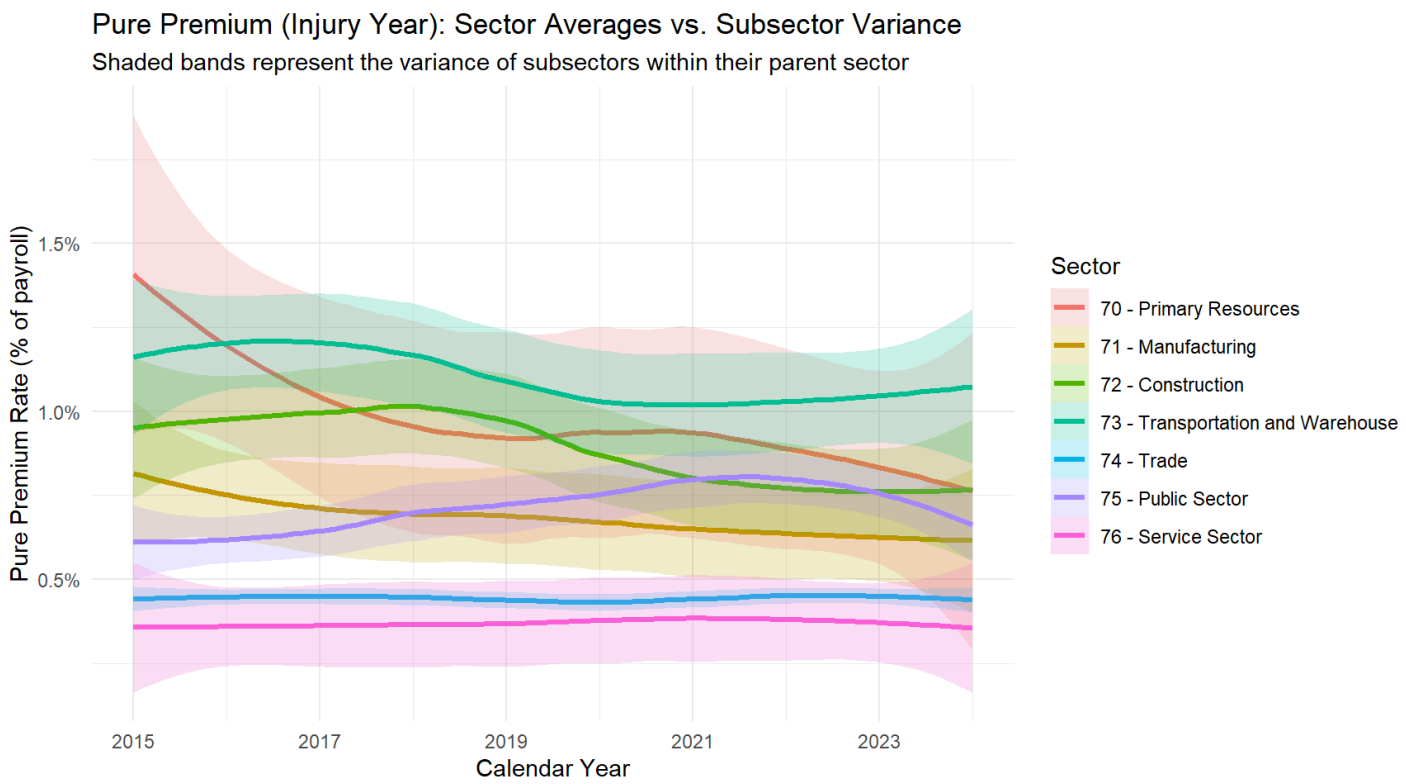
Facts:
claim cost = NA when CU never first exists
payroll = 0 always have claim_cost_iy = 0, not the way though.
753099 - Public Administration Combined CUs has no claim_cost_iy but positive claim_cost_oy
Only Subsector = 7020 - Fishing has person_years = NA
Only the following CUs have no payrolls -> person_years (One-to-one)
Only 722099 - Heavy Construction Combined CUs has no person_years but $10k payroll in 2016, 2017

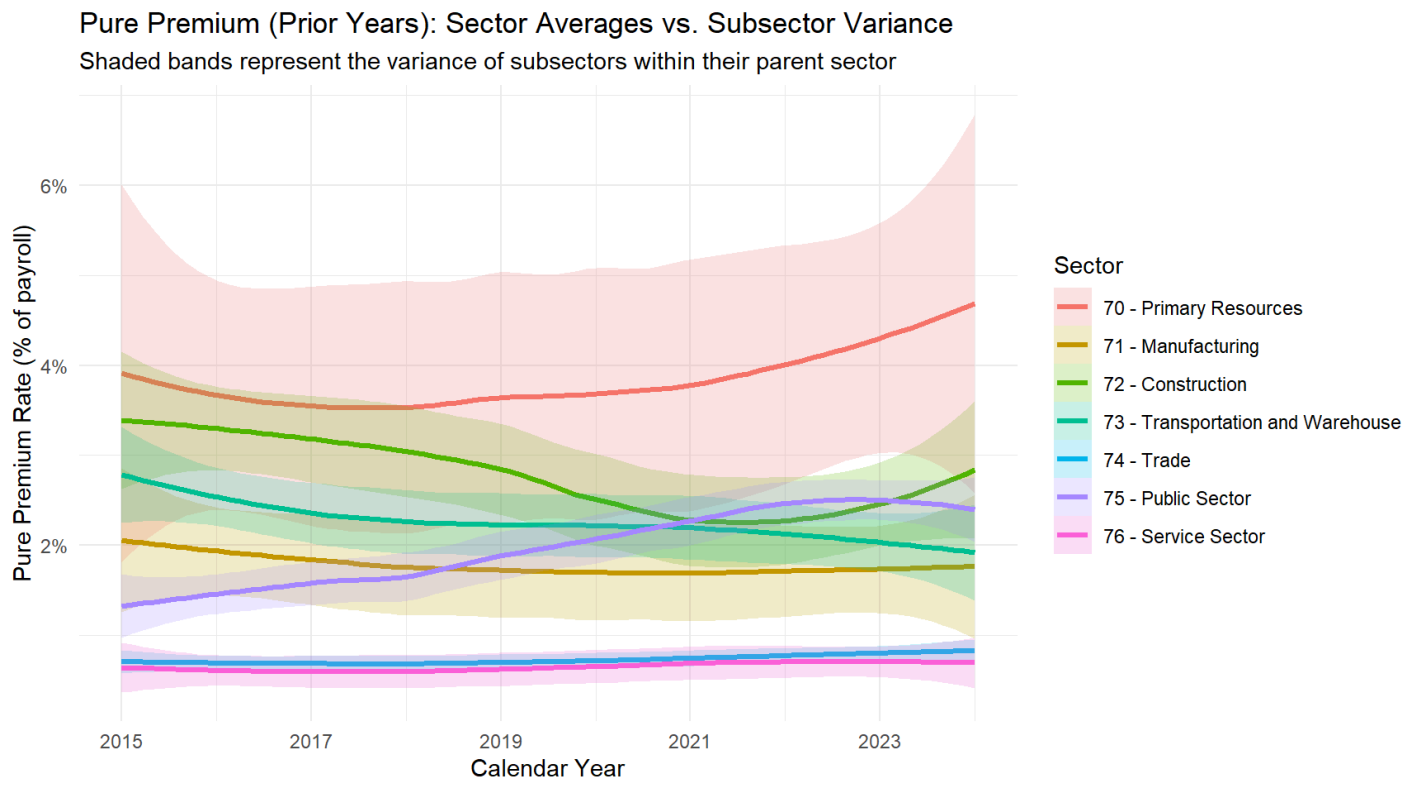
Subsector Modeling (Phase 1)

Subsector Model using Subsector aggregated data:

First Plots







Unified vs. Bifurcated Approach

Historically, it has been challenging to align current and prior claim costs with the correct exposure base. In this section, we test two distinct architectures:

The Single Model: A traditional approach attempting to predict total claim costs using the current year's payroll.

The Bifurcated Model: Splitting the risk into two distinct streams: Injury Year Risk (immediate front-end costs) and Runoff Risk (prior year development).

By isolating these timelines, we hypothesize that applying a lagged payroll exposure to the Runoff Risk will significantly reduce prediction error.

Phase 1 Modeling

Single Model

Fixed Model:

Mixed Model:

We got AIC: 6322.5 in Fixed Model, and AIC: 6262.4 in Mixed Model that has more degree of freedom on residuals.

Split Models

Current year First Paid Claims

Fixed Model:

Mixed Model:

We got AIC: 5895 in Fixed Model and AIC: 5893.6 in Mixed Model.

Prior Claim Models

0. Current-Year payroll

current year payroll as the exposure base does not align prior years' claims, instead we should match with mix of prior year's payrolls.

1a) 1-Year Lag payroll (Fixed Model)

AIC: 5448.7, Year_index = 0.006559 with p-value 0.243.

1b) 1-Year Lag payroll (Mixed Model)

AIC: 5418.4, Year_index = 0.005299 with group variance 0.003276, Var(RES) = 0.0322

1c) 2-Year Lag payroll (Mixed Model)

AIC: 4670.6, Year_index = 0.02396 with group variance 0.004379, Var(RES) = 0.0352. AIC reduces as observations reduce, not a fair comparison, we should compare with RMSPE.

Validation Step

Testing the Lag Hypothesis
To mathematically prove the optimal exposure base for long-tail runoff, we instituted an 8:2 train/test split. The models were trained on data up to 2022, and their predictive accuracy (RMSPE) was evaluated against the unseen 2023-2024 testing data.

```
## [1] "RMSPE of Single Fixed Model: $ 12,481,311.02"
```

```
## [1] "RMSPE of Single Mixed Model: $ 8,861,863.64"
```

```
## [1] "RMSPE of Bifurcated Fixed Model of 1 Year Prior: $ 13,887,321.28"
```

```
## [1] "RMSPE of Bifurcated Mixed Model of 0 Years Prior: $ 8,065,181.80"
```

```
## [1] "RMSPE of Bifurcated Mixed Model of 1 Year Prior: $ 6,847,133.55"
```

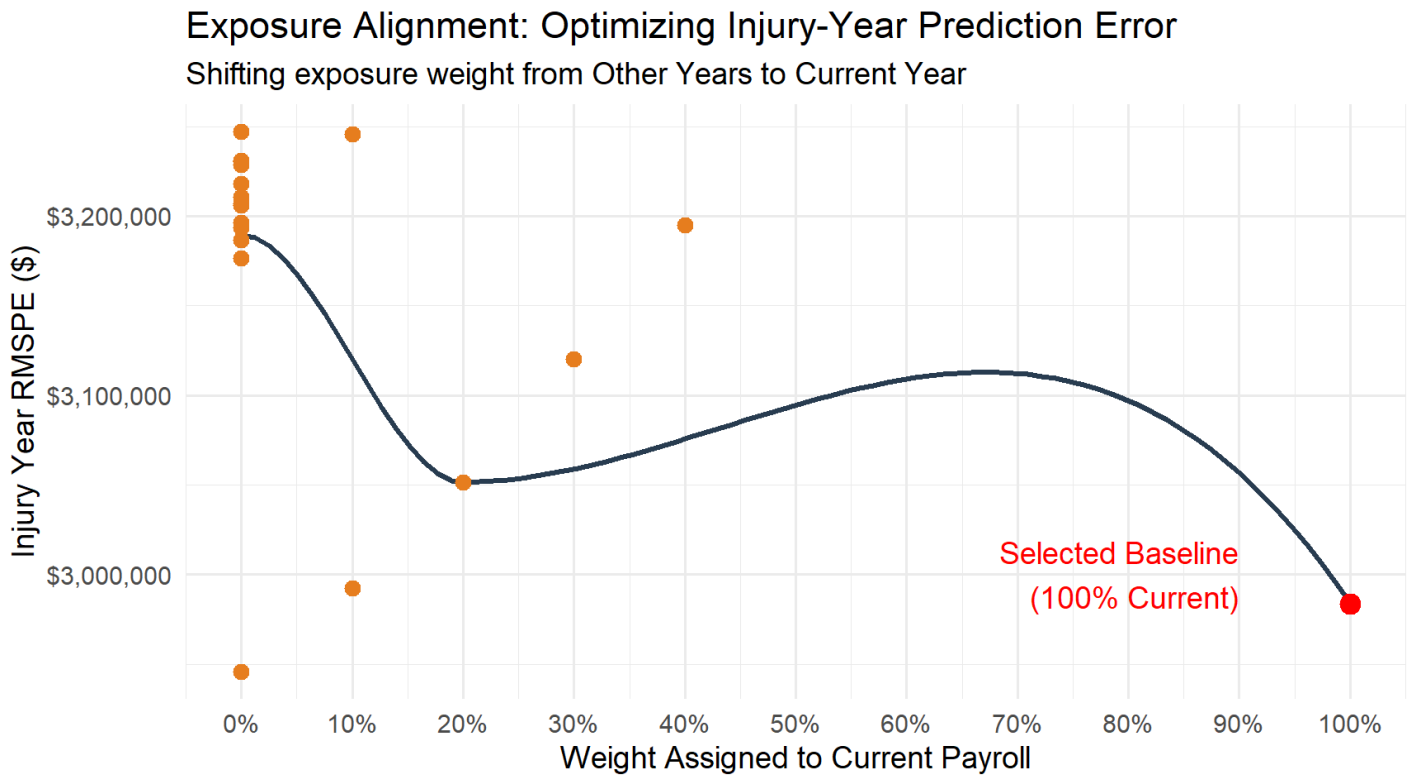
```
## [1] "RMSPE of Bifurcated Mixed Model of 2 Years Prior: $ 8,013,351.26"
```

The bifurcated Lag 1 architecture outperformed the Lag 0 approach, reducing the overall prediction error by nearly $1.2 million.
We further utilized a grid search algorithm to test various weighted moving averages of historical payroll (Lags 1 through 4) to see if deeper historical smoothing outperformed a simple 1-year lag.

To get the optimized weights

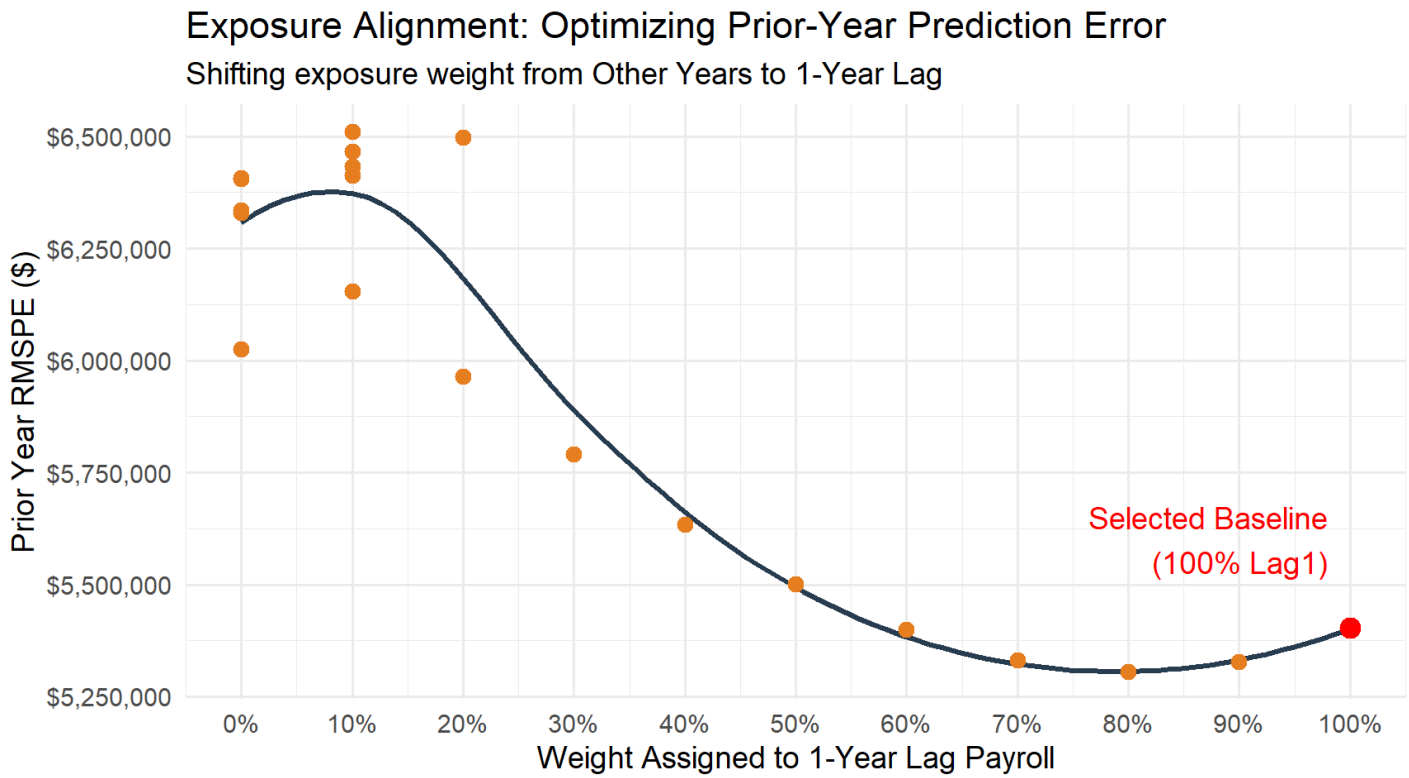
(# Weights <= 4 for 8:2 split)

```
##    w0  w1 w2            w3 RMSPE_iy
## 1 0.0 1.0  0  0.000000e+00  2945535
## 2 1.0 0.0  0  0.000000e+00  2983596
## 3 0.1 0.9  0  0.000000e+00  2992286
## 4 0.2 0.8  0  0.000000e+00  3051485
## 5 0.3 0.7  0 -1.110223e-16  3119951
```

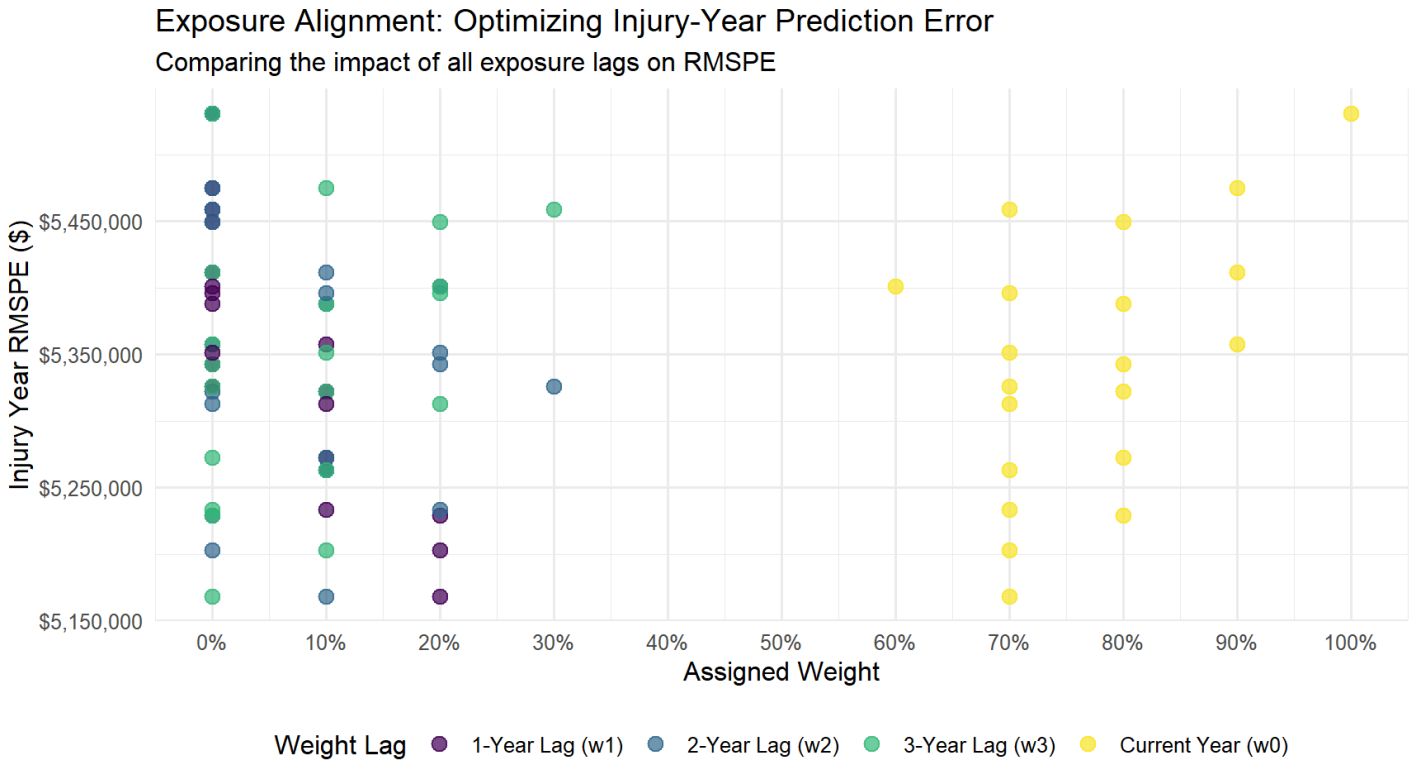


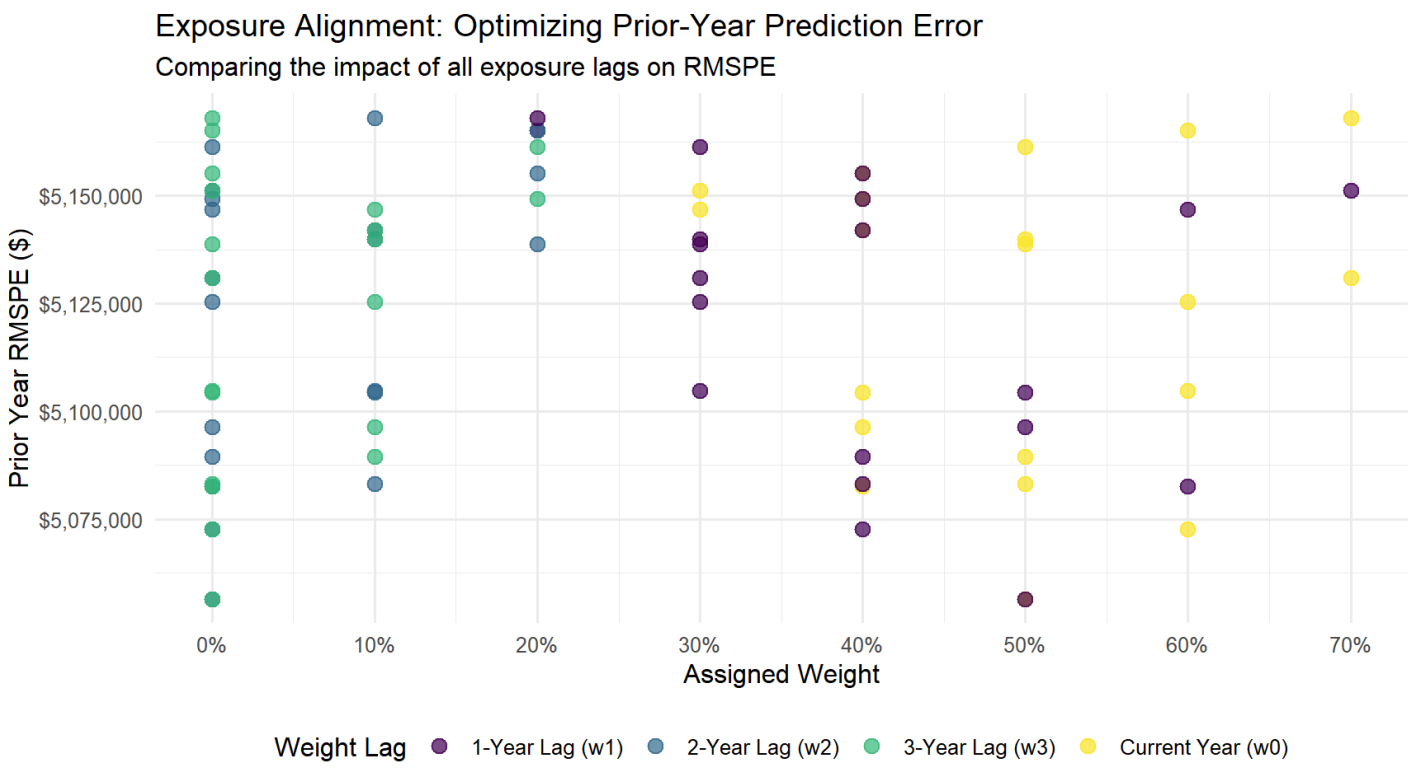
Yes, lag1 is overfitting on test years… To be honest, we couldn't figure out a better way to adjust this but don't forget here we aim to detrend and we are running out of degree of freedom (Otherwise we could put all available lags as offsets!). At least from below we found lag1 generalize most of the claim costs from past decade therefore it is better suit on the runoff costs. Now we do the same with OY,

```
##    w0  w1 w2            w3 RMSPE_iy RMSPE_oy
## 1 0.2 0.8  0  0.000000e+00  3051485  5304956
## 2 0.1 0.9  0  0.000000e+00  2992286  5327206
## 3 0.3 0.7  0 -1.110223e-16  3119951  5331002
## 4 0.4 0.6  0 -1.110223e-16  3194816  5398792
## 5 0.0 1.0  0  0.000000e+00  2945535  5403901
```



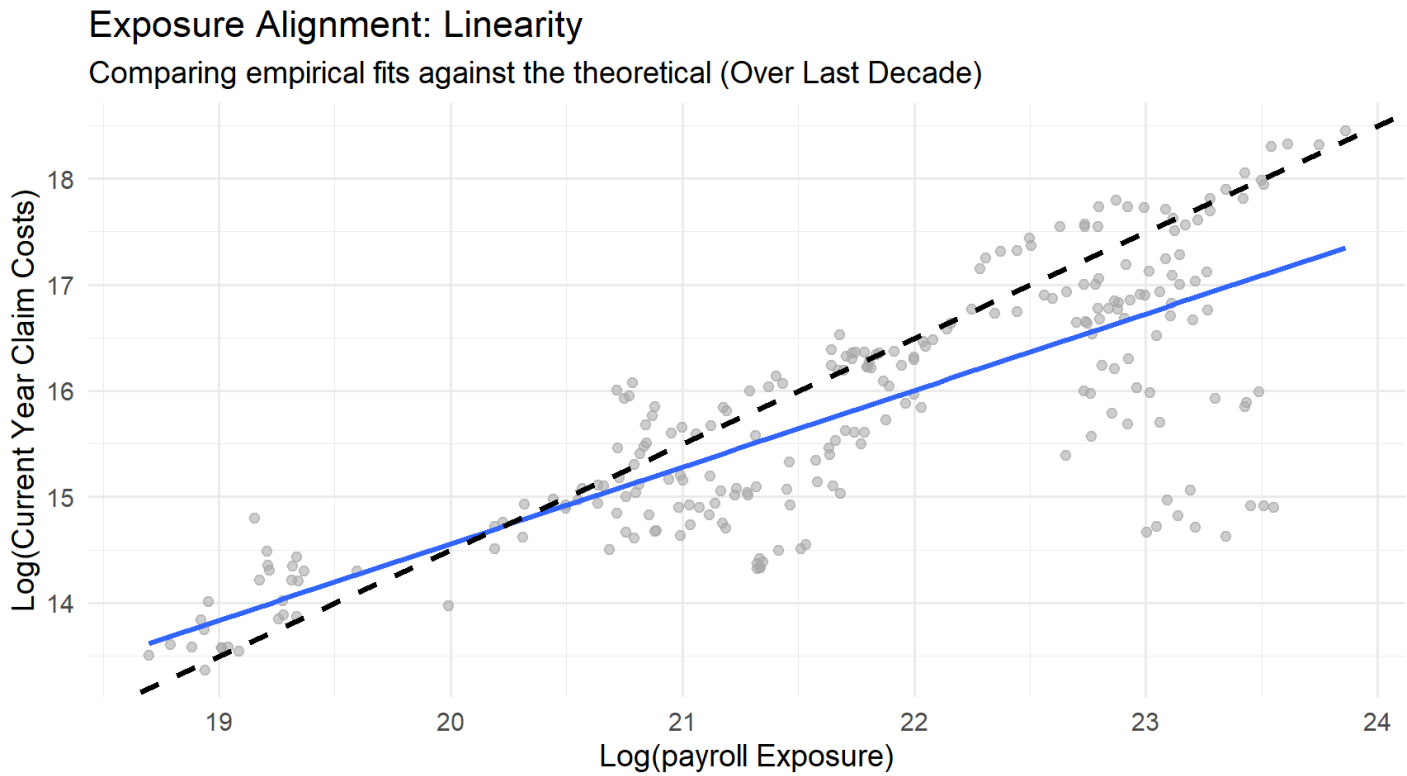
Post-report: This could be a less bias way to run the weight grid, 3:1 to 6:1 validation test. This suggests a 50:50 on lag1 and current on prior costs. Note that the proportion would be different if we drop w3 and extend the testing window by one more year, that's why we don't want to put too much faith on such results, knowing the alignment on lag1 is enough.

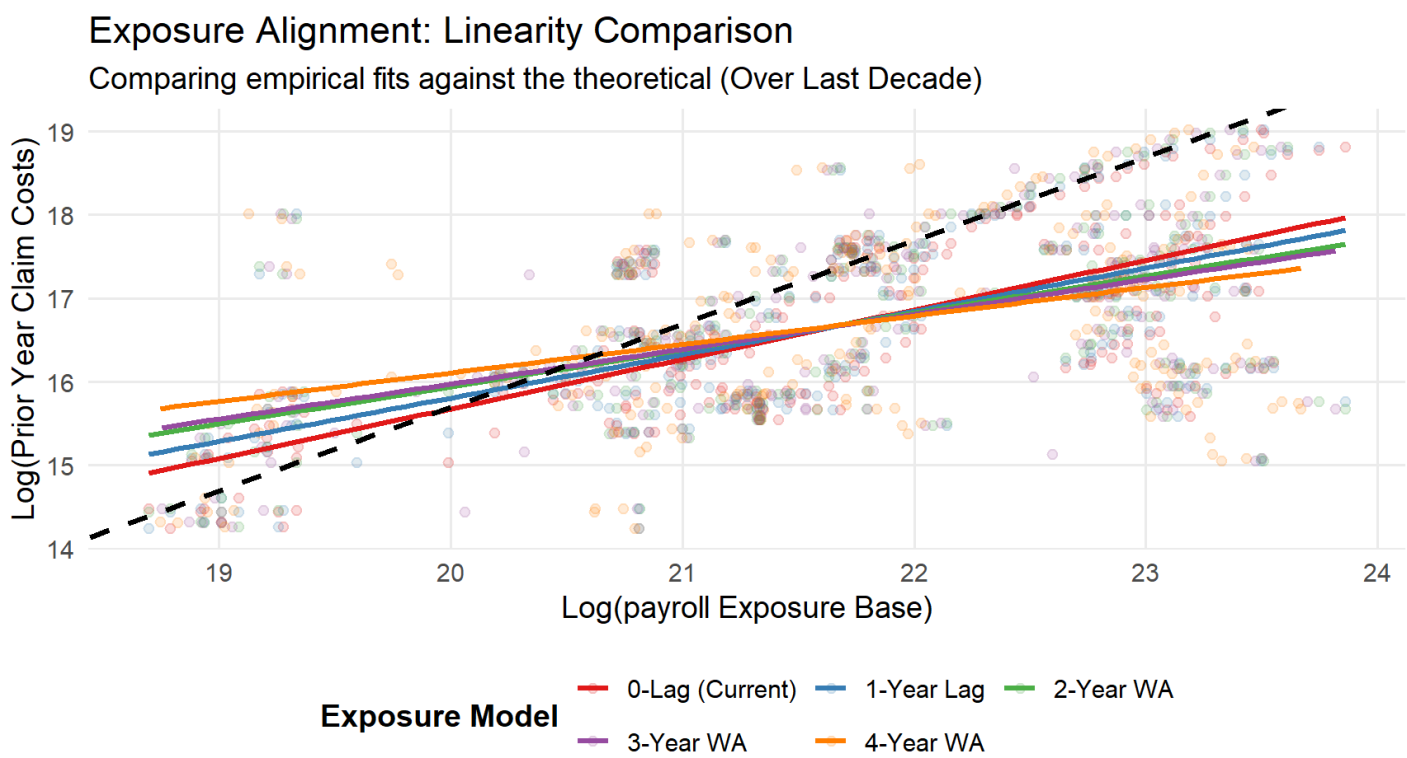




The optimization confirmed that placing high weight on the 1-year lagged payroll minimizes the Root Mean Square Prediction Error (RMSPE), you see some lag0 signals could be more favorable, but for simplicity we put 100% on 1-year lag. Also deeper historical lags or complex moving averages did not improve accuracy, indicating that the immediate past-year workforce volume is the single strongest predictor of near-term runoff cash flow.

Weighted Exposures Linearity:





The log-log linearity of payroll base vs claims costs holds, but there exists some high exposure subsectors having lower pure premiums.

Financial Validation

Error Margin & Premium Absorption:

```
## [1] "Annualized Provincial Error: $ 164,331,205.31"
```

```
## [1] "Average Annual Claim Costs: $ 1,389,328,146.50"
```

```
## [1] "Average Annual Premium Base: $ 2,358,256,050.50"
```

```
## [1] "--> Claim Cost Error Margin: 11.83 %"
```

```
## [1] "--> Premium Absorption Rate: 6.97 %"
```

While the methodology is sound, this 11.83% error margin suggests that significant variances remain hidden within specific industries. To address this, we must transition from Subsector-level models down to granular, CU-level analyses.

Full 10-Year Subsector Trends

Using the current best-of-fit model to analyze risks over a decade

To compare the models, we do 8 year (2017 - 2024) evaluation.

Final Model Evaluation Summary

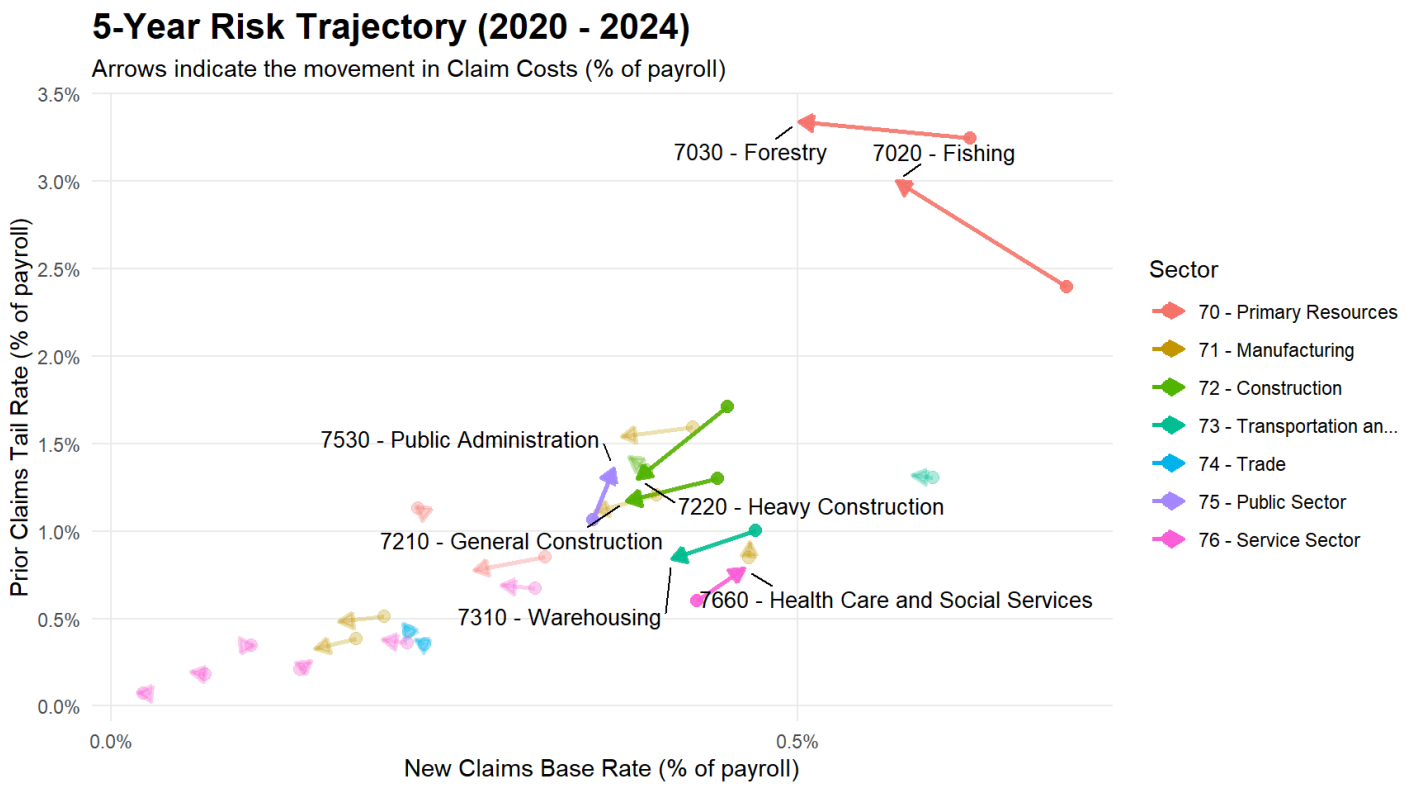
Architecture	Model Type	Exposure Base	8-Year Fit (RMSE)	Test Set (RMSPE)	Claim Cost Error Margin
Single	Fixed	Lag 0	$8,022,232	$12,481,311	21.56%
Single	Mixed	Lag 0	$4,419,811	$8,861,864	15.31%
Bifurcated	Fixed	IY: Lag 0 / OY: Lag 1	$5,162,637	$13,887,321	23.99%
Bifurcated	Mixed	IY: Lag 0 / OY: Lag 0	$4,527,384	$8,065,182	13.93%
Bifurcated	**Mixed**	**IY: Lag 0 / OY: Lag 1**	**$4,284,464**	**$6,847,134**	**11.83%**
Bifurcated	Mixed	IY: Lag 0 / OY: Lag 2	$4,430,322	$8,013,351	13.84%

Phase 2 Modeling

Subsector Model using CU level data:

The full model with CU as random effects is very difficult to interpret. In order to showcase a presentable results, we aim at the outstanding subsectors and target their CUs for further structural risk analysis.

Random Effects



Top Risk Trajectories by Subsector

Subsector	trend_IY	trend_OY	Vector_Length	Rate_Drift
7020 - Fishing	-4.8%	5.8%	0.62%	0.48%
7220 - Heavy Construction	-3.9%	-6.7%	0.42%	0.48%
7530 - Public Administration	1.1%	6.3%	0.29%	0.31%
7660 - Health Care and Social Services	1.9%	6.9%	0.19%	0.22%
7310 - Warehousing	-3.4%	-4.3%	0.17%	0.22%
7030 - Forestry	-5.4%	0.7%	0.16%	0.03%
7210 - General Construction	-3.9%	-2.6%	0.14%	0.19%

CU level Modeling

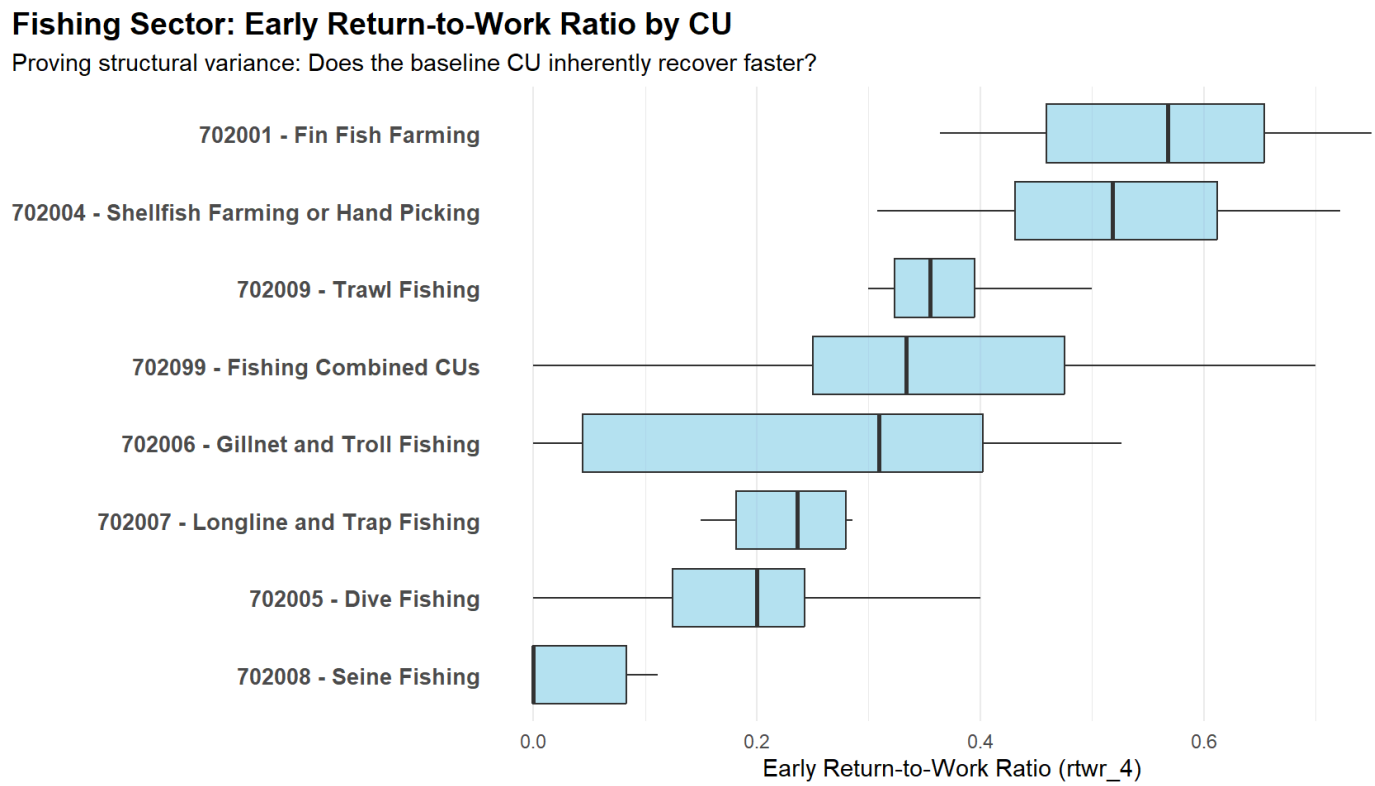
1. 7020 - Fishing

```
##
## Fishing: Stepwise GLM Selection
## =======================================================
##                              Dependent variable:
##                       ---------------------------------
##                          Claim Costs     claim_cost_oy
##                       Injury Year (IY)  Prior Year (OY)
##                              (1)              (2)
## -------------------------------------------------------
## time_loss_claims      0.014* (0.008)   -0.028** (0.011)
## rtwr_4               -1.487*** (0.473)
## I(avg_days_lost/100)                    0.224 (0.151)
## Constant             0.519*** (0.178)  0.785** (0.358)
## -------------------------------------------------------
## Observations                 80                72
## Akaike Inf. Crit.        2,082.494        2,110.665
## =======================================================
## Note:                        *p<0.1; **p<0.05; ***p<0.01
```

```
##
## Fishing: Stepwise GLM Selection
## ==============================================================================
##                                                        Dependent variable:
##                                                 ----------------------------------
##                                                    Claim Costs      claim_cost_oy
##                                                 Injury Year (IY)   Prior Year (OY)
##                                                        (1)               (2)
## ------------------------------------------------------------------------------
## time_loss_claims                                 0.021** (0.010)
## I(avg_days_lost/100)                                                0.148* (0.078)
## CU702004 - Shellfish Farming or Hand Picking 0.963*** (0.260)  0.659*** (0.214)
## CU702005 - Dive Fishing                      1.267*** (0.316)  1.108*** (0.243)
## CU702006 - Gillnet and Troll Fishing         1.280*** (0.278)  1.180*** (0.218)
## CU702007 - Longline and Trap Fishing         0.670*** (0.247)   0.463** (0.221)
## CU702008 - Seine Fishing                     2.064*** (0.316)  2.106*** (0.236)
## CU702009 - Trawl Fishing                     1.625*** (0.254)  1.546*** (0.217)
## CU702099 - Fishing Combined CUs              1.576*** (0.320)  2.627*** (0.228)
## Constant                                    -1.356*** (0.317) -1.067*** (0.161)
## ------------------------------------------------------------------------------
## Observations                                        80                72
## Akaike Inf. Crit.                                2,044.746         2,021.517
## ==============================================================================
## Note:                                                 *p<0.1; **p<0.05; ***p<0.01
```

Baseline: CU702001 - Fin Fish Farming

How the CU replaced rtwr_4?



Pick Significant Cost Drivers:
IY: time_loss_claims
OY: avg_days_lost

Mixed Model with selected drivers:

Fishing: GLMM Cost Drivers

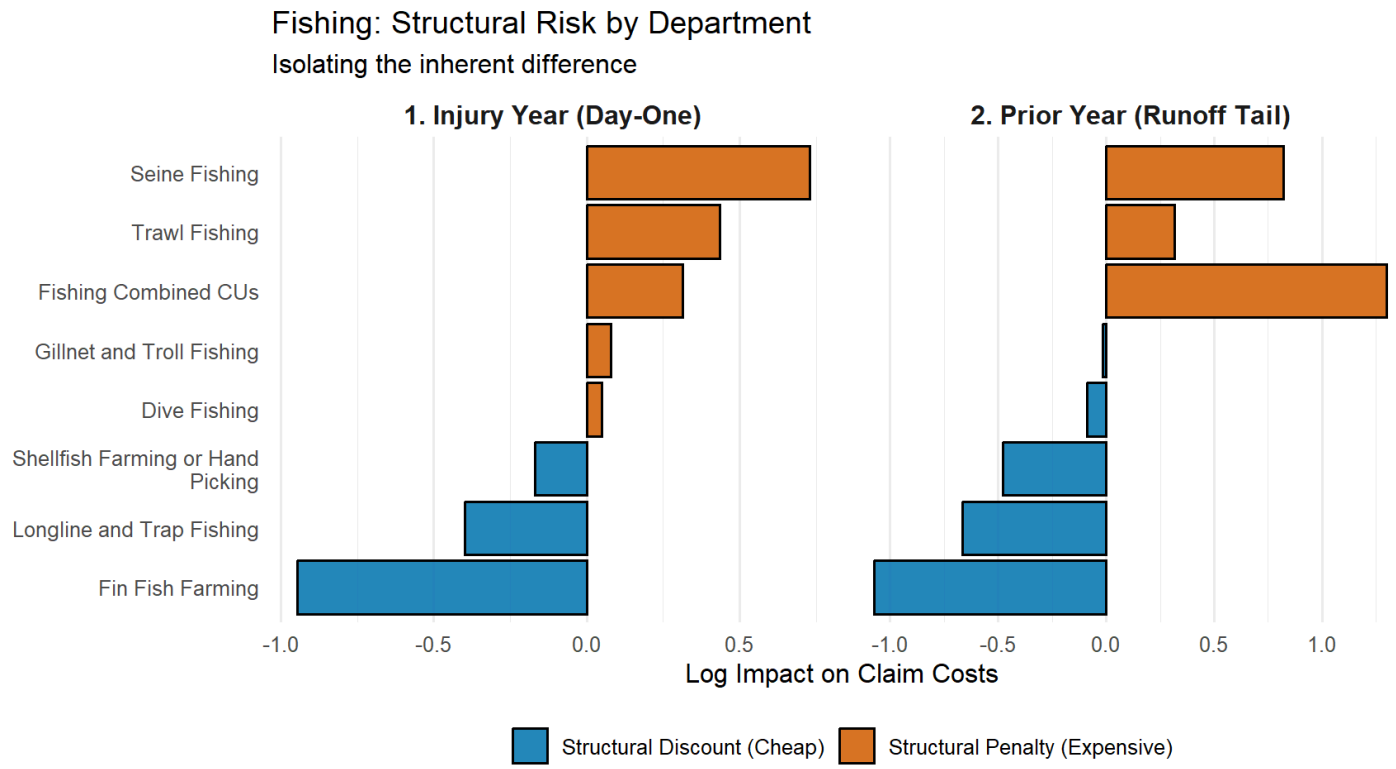
	Injury Year (IY)			Prior Year (OY)		
Characteristic	**exp(Beta)**	**95% CI**	**p-value**	**exp(Beta)**	**95% CI**	**p-value**
(Intercept)	0.886	0.511, 1.54	0.7	1.12	0.576, 2.18	0.7
I(time_loss_claims/10)	1.18	1.00, 1.40	**0.046**			
I(avg_days_lost/100)				1.17	1.01, 1.36	**0.039**
Abbreviation: CI = Confidence Interval						

The intercept p-value is large, meaning the priori offset is doing its job.
Frequency trend is detected in injury year and Severity trend is detected in prior year.

Injury Year:
Holding the structural risk of the boat constant, every additional time-loss claim acts as a 1.02x multiplier on day-one costs.
Prior Year:
Because avg_days_lost was scaled, one sd increase in the average claim duration acts as a 1.15x multiplier on driving up the long-tail runoff costs.

Now we move onto the remaining variances captured by the CUs.

The CU deviance:



CU-level Risk:
Fin Fish Farming is structurally insulated from both Day-One and long-tail risks (Random Intercepts consistently around -1). Their controlled environments naturally suppress severe injuries and easily accommodate light-duty work. Seine Fishing (IY: +0.79, OY: +0.62) and Trawl Fishing (IY: +0.36, OY: +0.53) are inherently dangerous. Their environments structurally dangerous and act as mathematical multipliers for catastrophic, long-tail claim deteriorations. They require entirely different premium pricing structures.

2. 7220 - Heavy Construction

```
##
## Heavy Construction: Stepwise GLM Selection
## ==========================================================================
##                                            Dependent variable:
##                              ---------------------------------------------
##                                 Claim Costs    pmax(1, claim_cost_oy_smth)
##                              Injury Year (IY)        Prior Year (OY)
##                                     (1)                    (2)
## --------------------------------------------------------------------------
## time_loss_claims              0.028** (0.011)        0.013* (0.007)
## I(avg_wage/10000)             0.659** (0.295)       -0.134** (0.063)
## Post_2020                     4.360** (2.021)
## rtwr_4                       2.109*** (0.670)
## I(avg_wage/10000):Post_2020 -0.762** (0.323)
## avg_days_lost                                         0.002 (0.001)
## Constant                     -5.245*** (1.611)        0.613 (0.465)
## --------------------------------------------------------------------------
## Observations                        74                     66
## Akaike Inf. Crit.               1,882.071              1,883.376
## ==========================================================================
## Note:                                          *p<0.1; **p<0.05; ***p<0.01
```

```
##
## Heavy Construction: Stepwise GLM Selection
## ==========================================================================================
==========================================================
##
Dependent variable:
##
---------------------------------------------
##
Claim Costs    pmax(1, claim_cost_oy_smth)
##
Injury Year (IY)        Prior Year (OY)
##
(1)                     (2)
## ------------------------------------------------------------------------------------------
----------------------------------------------------------
## time_loss_claims
0.035** (0.013)       -0.014** (0.007)
## I(avg_wage/10000)
0.711** (0.296)
## Post_2020
3.564* (1.906)
## rtwr_4
1.562** (0.669)
## CU722002 - House Raising or Structural Moving of Buildings or Heavy Equipment
0.295 (0.380)          0.273 (0.227)
## CU722003 - Pier, Wharf, or Dry Dock Construction or Repair
-0.691* (0.391)        0.423* (0.239)
## CU722004 - Pile Driving or Dredging
-0.835* (0.418)       -0.705*** (0.223)
## CU722005 - Steel Frame Erection, Repair, Painting, or Bridge Painting or Cleaning
-0.179 (0.378)        0.724*** (0.226)
## CU722010 - Mechanized Tunnel Boring Construction
-0.722** (0.359)       -0.609*** (0.222)
## CU722011 - Structural Concrete Reservoir, Dam, Dyke, or Tunnel Construction or Repair (not elsewhe
re specified)  -0.685 (0.418)          0.130 (0.224)
## CU722099 - Heavy Construction Combined CUs
-0.883 (0.780)        -1.349*** (0.327)
## I(avg_wage/10000):Post_2020
-0.648** (0.304)
## Constant
-5.145*** (1.580)        0.211 (0.189)
## ------------------------------------------------------------------------------------------
----------------------------------------------------------
## Observations
74                     66
## Akaike Inf. Crit.
1,879.638              1,879.105
## ==========================================================================================
==========================================================
## Note:
*p<0.1; **p<0.05; ***p<0.01
```

Note: Warning comes from missing 2015 - 2020 data from 722099 - Heavy Construction Combined CUs, which can be safely ignored since the later GLMM will partially pool it to the average.

Baseline: CU722001 - Bridge, Overpass, or Viaduct Construction or Repair

Anova Test

```
## Data: hvyconst_clean
## Models:
## variance_hvyconst_iy_without_boom: claim_cost_iy ~ time_loss_claims + (1 | CU) + offset(logpay_iy_
trend), zi=~0, disp=~1
## variance_hvyconst_iy_without_rtw: claim_cost_iy ~ time_loss_claims + (I(avg_wage/10000) * Post_202
0) + , zi=~0, disp=~1
## variance_hvyconst_iy_without_rtw:     (1 | CU) + offset(logpay_iy_trend), zi=~0, disp=~1
## variance_hvyconst_iy_with_rtw: claim_cost_iy ~ time_loss_claims + rtwr_4 + (I(avg_wage/10000) * ,
zi=~0, disp=~1
## variance_hvyconst_iy_with_rtw:     Post_2020) + (1 | CU) + offset(logpay_iy_trend), zi=~0, disp=~1
##                                   Df    AIC    BIC  logLik deviance  Chisq
## variance_hvyconst_iy_without_boom  5 1849.2 1860.7 -919.58   1839.2
## variance_hvyconst_iy_without_rtw   8 1852.3 1870.8 -918.17   1836.3 2.8111
## variance_hvyconst_iy_with_rtw      9 1850.4 1871.2 -916.21   1832.4 3.9244
##                                   Chi Df Pr(>Chisq)
## variance_hvyconst_iy_without_boom
## variance_hvyconst_iy_without_rtw       3    0.42168
## variance_hvyconst_iy_with_rtw          1    0.04759 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Data: hvyconst_clean
## Models:
## variance_hvyconst_iy_without_boom: claim_cost_iy ~ time_loss_claims + (1 | CU) + offset(logpay_iy_
trend), zi=~0, disp=~1
## variance_hvyconst_iy_with_rtw: claim_cost_iy ~ time_loss_claims + rtwr_4 + (I(avg_wage/10000) * ,
zi=~0, disp=~1
## variance_hvyconst_iy_with_rtw:     Post_2020) + (1 | CU) + offset(logpay_iy_trend), zi=~0, disp=~1
##                                   Df    AIC    BIC  logLik deviance  Chisq
## variance_hvyconst_iy_without_boom  5 1849.2 1860.7 -919.58   1839.2
## variance_hvyconst_iy_with_rtw      9 1850.4 1871.2 -916.21   1832.4 6.7355
##                                   Chi Df Pr(>Chisq)
## variance_hvyconst_iy_without_boom
## variance_hvyconst_iy_with_rtw          4     0.1505
```

```
##     psi
## 1.49115
```

```
##      psi
## 1.499121
```

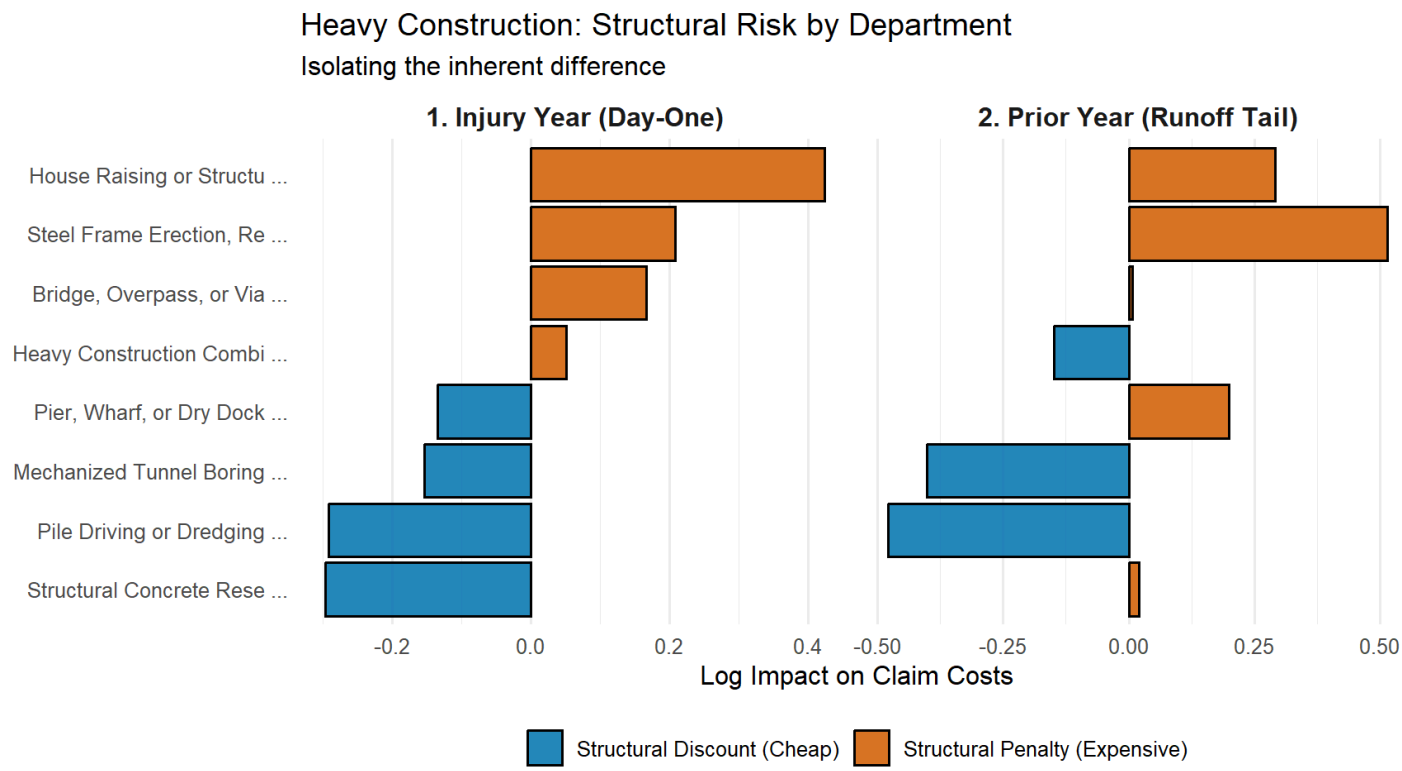
```
## Data: hvyconst_clean %>% filter(Year_index > 0)
## Models:
## variance_hvyconst_oy_without_tlc: claim_cost_oy_smth ~ (1 | CU) + offset(logpay_oy_trend), zi=~0,
disp=~1
## variance_hvyconst_oy_with_tlc: claim_cost_oy_smth ~ time_loss_claims + (1 | CU) + offset(logpay_oy
_trend), zi=~0, disp=~1
##                                  Df    AIC    BIC  logLik deviance  Chisq
## variance_hvyconst_oy_without_tlc  4 1821.7 1830.5 -906.85   1813.7
## variance_hvyconst_oy_with_tlc     5 1822.2 1833.2 -906.13   1812.2 1.4508
##                                  Chi Df Pr(>Chisq)
## variance_hvyconst_oy_without_tlc
## variance_hvyconst_oy_with_tlc         1     0.2284
```

Drop drop drop

Heavy Construction: GLMM Cost Drivers

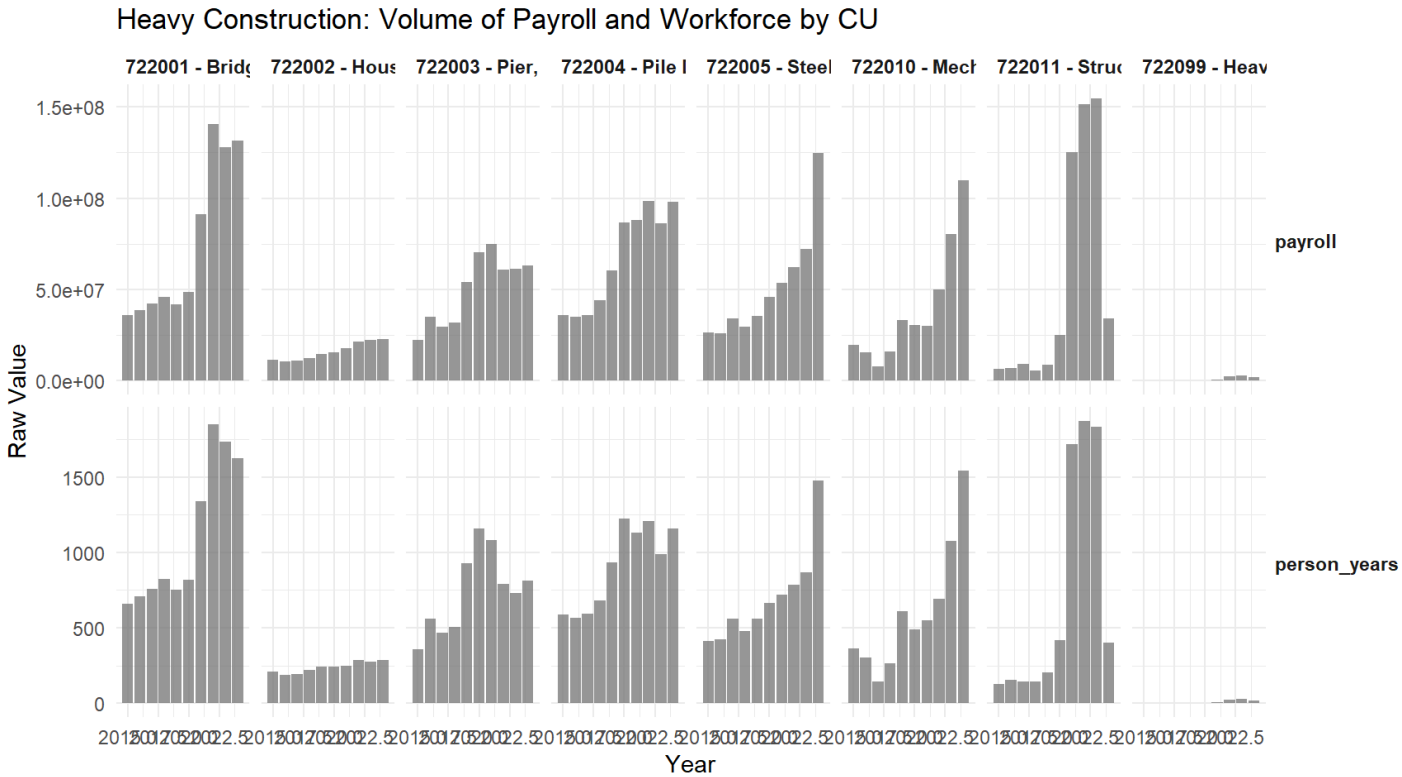
	Injury Year (IY)				Prior Year (OY)			
Characteristic	**Group**	**exp(Beta)**	**95% CI**	**p-value**	**Group**	**exp(Beta)**	**95% CI**	**p-value**
(Intercept)	cond	0.126	0.003, 6.10	0.3	cond	0.980	0.719, 1.34	0.9
time_loss_claims	cond	1.04	1.02, 1.06	**<0.001**				
I(avg_wage/10000)	cond	1.27	0.654, 2.48	0.5				
Post_2020	cond	3.66	0.060, 225	0.5				
I(avg_wage/10000) * Post_2020	cond	0.774	0.395, 1.52	0.5				

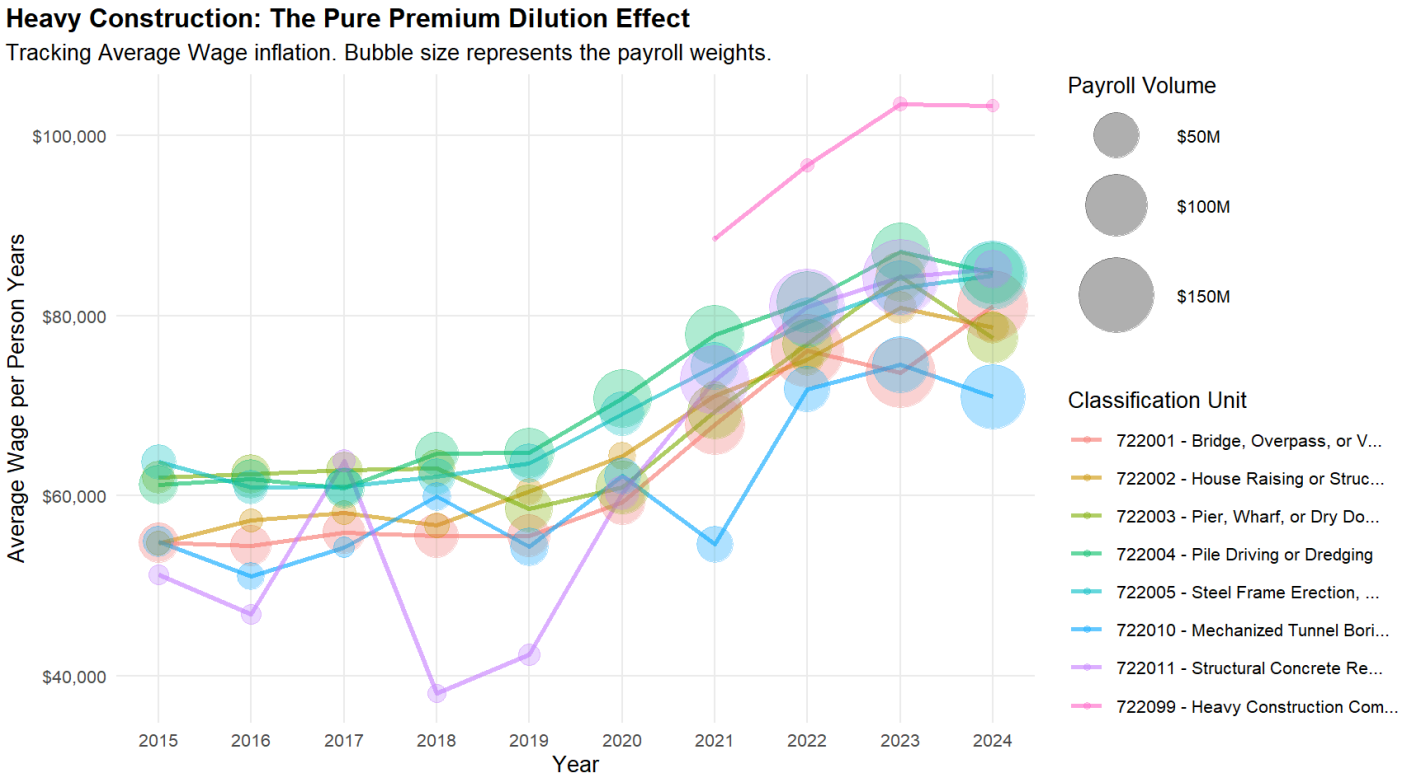
Abbreviation: CI = Confidence Interval



Heavy Construction Subsector Drivers:

IY:
First-year costs are heavily dictated by raw injury frequency (time_loss_claims acts as a highly significant 1.04x multiplier).
OY:
In the long-term liability, frequency metrics lose their predictive power, the cost is entirely hard-coded into the physical nature of the specific industry.





Highlight that heavy infrastructure CUs saw explosive growth post-2020 due to provincial mega-projects. Recall we have engineered the average wage metric (avg_wage) to test if high-exposure mega-projects carried inherently different risk rates. However, the stepwise algorithm rejected it. This mathematically proves two things: First, our 10-year baseline and CU variables successfully absorbed the structural differences from the mega-projects. Second, besides the RTW < 4 weeks effect on the first year cost, the risk is mostly tied to the machinery and the trade.

3. 7530 - Public Administration

```
##
## Public Admin: Stepwise GLM Selection
## ====================================================
##                           Dependent variable:
##                   -----------------------------------
##                      Claim Costs       claim_cost_oy
##                    Injury Year (IY)   Prior Year (OY)
##                          (1)                (2)
## ----------------------------------------------------
## time_loss_claims  0.001*** (0.00005) 0.001*** (0.0001)
## avg_days_lost                         0.007* (0.004)
## rtwr_26                               3.157* (1.664)
## aov_rate           3.374*** (0.203)  4.206*** (0.372)
## lrss_rate                            -3.820*** (1.137)
## violate                               0.429** (0.151)
## Constant          -0.973*** (0.063)  -1.298*** (0.281)
## ----------------------------------------------------
## Observations              30                27
## Akaike Inf. Crit.      868.553           849.383
## ====================================================
## Note:                      *p<0.1; **p<0.05; ***p<0.01
```

```
##
## Public Admin: Stepwise GLM Selection
## ==================================================================================
##                                                        Dependent variable:
##                                                  ---------------------------------
##                                                     Claim Costs     claim_cost_oy
##                                                  Injury Year (IY)  Prior Year (OY)
##                                                        (1)              (2)
## ----------------------------------------------------------------------------------
## time_loss_claims                                 0.0004* (0.0002)  0.001** (0.0003)
## log_avg_wage                                                        -0.645 (0.438)
## avg_days_lost                                                      0.013*** (0.004)
## aov_rate                                                            1.801 (1.186)
## lrss_rate                                                          -2.418** (1.118)
## violate                                                             0.291* (0.140)
## CU753003 - Law Enforcement                       1.177*** (0.063)  1.340*** (0.414)
## CU753004 - Local Government and Related Operations   0.128 (0.338)    0.073 (0.455)
## Constant                                        -0.824*** (0.062)  5.497 (4.528)
## ----------------------------------------------------------------------------------
## Observations                                            30                27
## Akaike Inf. Crit.                                    862.038           844.675
## ==================================================================================
## Note:                                                   *p<0.1; **p<0.05; ***p<0.01
```

Baseline: CU753002 - First Nations Operations

After introduced the CUs, the signal from the injury frequency (time_loss_claims) has weaken, mostly driven by Law Enforcement. The OY model may have better explanatory power from the injury severity drivers (avg_days_lost and

lrss_rate). Our proud aov_rate factor does not work after all, the Law Enforcement stole its significance.
Pick Significant Cost Drivers:
IY: time_loss_claims, time_loss_claims
OY: time_loss_claims, avg_days_lost, lrss_rate

Mixed Model with selected drivers:
We first fine-tune the model by deciding whether time_loss_claims needs to be dropped:

```
## Data: pbadmin_clean
## Models:
## variance_pbadmin_iy_without_tlc: claim_cost_iy ~ (1 | CU) + offset(logpay_iy_trend)
## variance_pbadmin_iy_with_tlc: claim_cost_iy ~ I(time_loss_claims/100) + (1 | CU) + offset(logpay_i
y_trend)
##                                  npar    AIC    BIC  logLik -2*log(L)  Chisq Df
## variance_pbadmin_iy_without_tlc    3 874.43 878.63 -434.21    868.43
## variance_pbadmin_iy_with_tlc       4 872.57 878.18 -432.29    864.57 3.8544  1
##                                  Pr(>Chisq)
## variance_pbadmin_iy_without_tlc
## variance_pbadmin_iy_with_tlc       0.04961 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Data: pbadmin_clean %>% filter(Year_index > 0)
## Models:
## variance_pbadmin_oy_without_tlclrss: claim_cost_oy ~ I(avg_days_lost/10) + (1 | CU) + offset(logpa
y_oy_trend)
## variance_pbadmin_oy_without_tlc: claim_cost_oy ~ I(avg_days_lost/10) + lrss_rate + (1 | CU) + offs
et(logpay_oy_trend)
## variance_pbadmin_oy_without_lrss: claim_cost_oy ~ I(time_loss_claims/100) + I(avg_days_lost/10) +
(1 | CU) + offset(logpay_oy_trend)
## variance_pbadmin_oy_with_tlclrss: claim_cost_oy ~ I(time_loss_claims/100) + I(avg_days_lost/10) +
lrss_rate + (1 | CU) + offset(logpay_oy_trend)
##                                     npar    AIC    BIC  logLik -2*log(L)  Chisq
## variance_pbadmin_oy_without_tlclrss    4 859.80 864.99 -425.90    851.80
## variance_pbadmin_oy_without_tlc        5 858.42 864.90 -424.21    848.42 3.3801
## variance_pbadmin_oy_without_lrss       5 861.41 867.89 -425.71    851.41 0.0000
## variance_pbadmin_oy_with_tlclrss       6 858.21 865.99 -423.11    846.21 5.2008
##                                     Df Pr(>Chisq)
## variance_pbadmin_oy_without_tlclrss
## variance_pbadmin_oy_without_tlc      1    0.06599 .
## variance_pbadmin_oy_without_lrss     0
## variance_pbadmin_oy_with_tlclrss     1    0.02258 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Data: pbadmin_clean %>% filter(Year_index > 0)
## Models:
## variance_pbadmin_oy_without_tlclrss: claim_cost_oy ~ I(avg_days_lost/10) + (1 | CU) + offset(logpa
y_oy_trend)
## variance_pbadmin_oy_with_tlclrss: claim_cost_oy ~ I(time_loss_claims/100) + I(avg_days_lost/10) +
lrss_rate + (1 | CU) + offset(logpay_oy_trend)
##                                    npar    AIC    BIC  logLik -2*log(L)  Chisq
## variance_pbadmin_oy_without_tlclrss    4 859.80 864.99 -425.90    851.80
## variance_pbadmin_oy_with_tlclrss       6 858.21 865.99 -423.11    846.21 5.5944
##                                    Df Pr(>Chisq)
## variance_pbadmin_oy_without_tlclrss
## variance_pbadmin_oy_with_tlclrss     2    0.06098 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

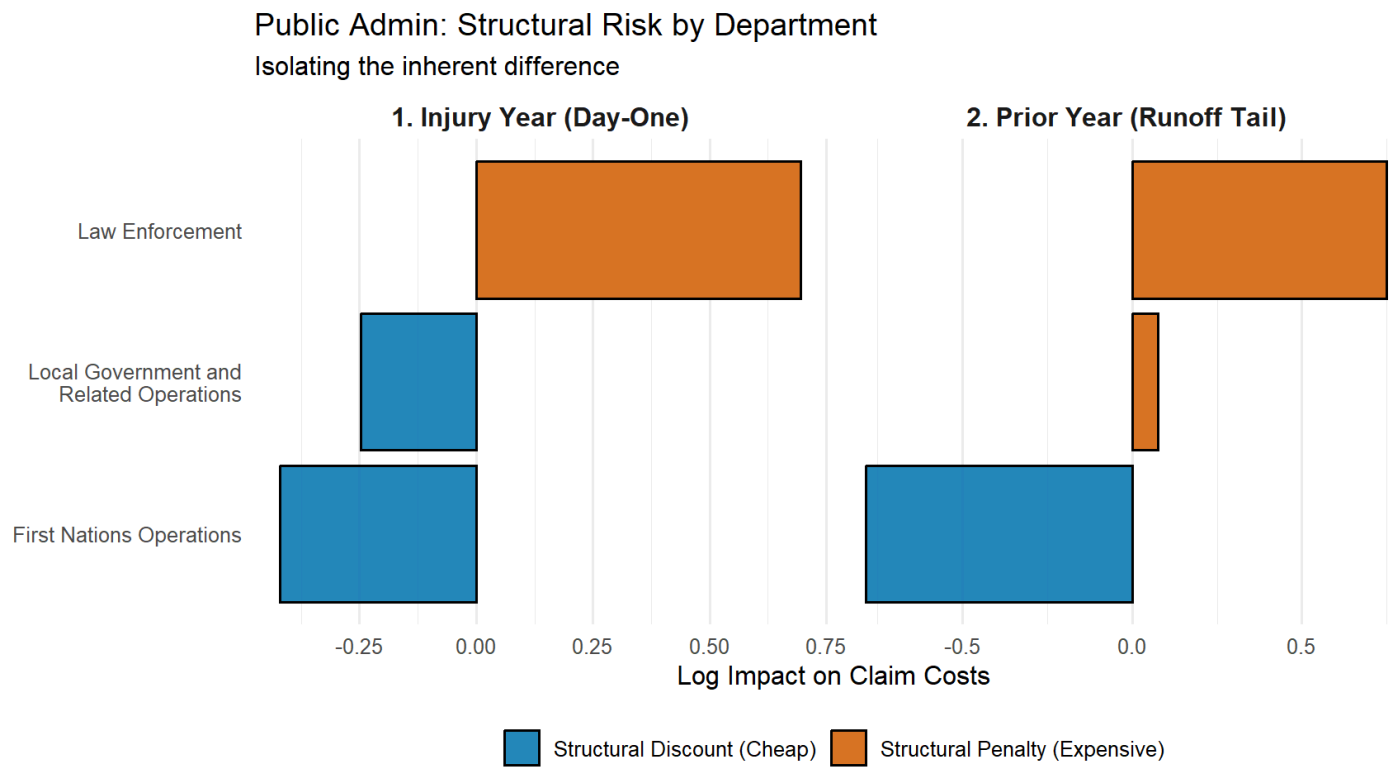
The time_loss_claims beats the reduced IY model but lost in the OY model.
Pick Significant Cost Drivers (Final):
IY: time_loss_claims
OY: avg_days_lost, lrss_rate

Public Admin: GLMM Cost Drivers

	Injury Year (IY)			Prior Year (OY)		
Characteristic	**exp(Beta)**	**95% CI**	**p-value**	**exp(Beta)**	**95% CI**	**p-value**
(Intercept)	0.689	0.304, 1.56	0.4	0.682	0.247, 1.89	0.5
I(time_loss_claims/100)	1.04	1.00, 1.08	**0.047**			
I(avg_days_lost/10)				1.07	1.00, 1.14	**0.047**

Abbreviation: CI = Confidence Interval

The CU deviance:



As proven by the Likelihood Ratio Test (ANOVA with p = 0.04961, 95% reject = 0), the overall claim frequency (time_loss_claims) becomes a significant driver of immediate costs across the entire sector (+1.34(scaled), p < 0.047). Regardless of whether the injury happens in a local government office or on police patrol, more injuries universally mean higher immediate medical and wage-loss expenses.

In the runoff tail, the claim frequency loses predictive power (ANOVA with p = 0.137, not able to reject = 0), meaning the claim frequency has no bearing on long-tail liability. Instead, severity—measured by the average claim duration (scale(avg_days_lost)) becomes the universal trigger, driving a 1.10x penalty multiplier on runoff costs.

Law Enforcement carries a massive structural penalty across both Injury Year and Prior Year models. When introducing the “Acts of Violence” (aov_rate) feature, the stepwise algorithm initially showed massive significance. However, once the CU variables were introduced, Law Enforcement completely absorbed the aov_rate’s significance. The algorithm cannot mathematically separate the concept of “Law Enforcement” from “extreme rates of violence. (multicollinearity)